

Rosenberg Compact-Operator Model

MATHLIBANNEX / SELECTED READING SCOPE

The mathematical goal

Let A be a nonzero complex C^* -algebra, not assumed unital, and let π act on a separable complex Hilbert space H as a representative of the unique unitary-equivalence class of nonzero irreducible representations of A . Then π is injective, every represented operator is compact, and every compact operator on H has an exact preimage in A .

33 approved Cards in this selected reading scope. The catalog contains 188 distinct Cards across the four Projects.

Scope

The result identifies A , through π , with the compact-operator model in an exact unbundled sense. The Project exposes the reusable mathematical stages through faithfulness, character eigenvectors, countability, scalar corners, rank-one preimages and compact approximation.

Hypotheses

A is nonzero and may be genuinely nonunital. H is separable. The representation π is nonzero and irreducible, and every nonzero irreducible representation of A is unitarily equivalent to π . No separability of A , prior faithfulness or simplicity is assumed. The unital finite-dimensional consequences have their own exact hypotheses.

Scope limits

The 33 Card explanations have completed source-exposition correspondence review against exact MathlibAnnex v0.4.0 source. The compact-operator characterization is unbundled; no bundled StarAlgEquiv is supplied. The original 1953 paper and the Farah book remain unacquired and unreviewed in this record.

Mathematical routes

The unique nonzero irreducible class gives faithfulness. Characters and countability yield a rank-one image; exact preimages of rank-one operators and compact approximation supply one inclusion, and the preimage ideal supplies the other. The unital finite-dimensional consequences form a final separate stage.

Each route shows its direct Cards and the reused prerequisites needed in the selected reading graph. Shared Cards count once within each route and once in the Catalog.

Representations, irreducibility and equivalence

Representations without a unit equation, nonzero irreducibility and unitary equivalence set the language of the theorem. The singleton condition names one representative of the unique nonzero irreducible class.

4 direct Cards + 0 reused prerequisites = 4 unique Cards. Shared Cards count once in this route.

- Key definition: [A representative of the unique irreducible class](#)
- Key definition: [Representations without a unit-preservation requirement](#)

[Read this route with prerequisites](#)

Pure states and faithfulness

A pure state detects a hypothetical nonzero kernel element and its GNS representation is irreducible. Unitary equivalence to the singleton representative then contradicts that kernel element, giving faithfulness and the related simplicity consequence.

5 direct Cards + 5 reused prerequisites = 10 unique Cards. Shared Cards count once in this route.

- Key result: [Faithfulness from a unique irreducible class](#)
- Key result: [A faithful singleton model forces simplicity](#)

[Read this route with prerequisites](#)

Characters, eigenvectors and a scalar corner

Characters of a maximal abelian subalgebra of the unitization extend to pure states. A character distinct from the scalar character yields a joint unit eigenvector in the singleton model. Countability and an isolated non-scalar character then give a nonzero scalar-corner projection and its rank-one image.

11 direct Cards + 8 reused prerequisites = 19 unique Cards. Shared Cards count once in this route.

- Key result: [A character becomes a joint unit eigenvector](#)
- Key result: [A projection represented by a rank-one operator](#)

[Read this route with prerequisites](#)

Every compact operator has a preimage

One represented rank-one projection yields every rank-one operator using dense orbits. Closed-range and compact approximation arguments put all compact operators in the image of the original algebra.

3 direct Cards + 16 reused prerequisites = 19 unique Cards. Shared Cards count once in this route.

- Key result: [Compact operators lie in a singleton representation range](#)
- Key result: [Every rank-one operator has an algebra preimage](#)

[Read this route with prerequisites](#)

Every represented operator is compact

The preimage ideal of compact operators and a separating character exclude a noncompact represented image. The contradiction uses the same projection while forcing the character to take both zero and one.

7 direct Cards + 14 reused prerequisites = 21 unique Cards. Shared Cards count once in this route.

- Key result: [Every operator in a singleton image is compact](#)
- Key definition: [The preimage ideal of the compact operators](#)

[Read this route with prerequisites](#)

Compact-operator models and unital consequences

Faithfulness and both compact-image inclusions identify the nonunital compact-operator model in an unbundled sense. In the unital singleton case the identity becomes compact, yielding finite dimension and the exact full-operator image consequences.

12 direct Cards + 20 reused prerequisites = 32 unique Cards. Shared Cards count once in this route.

- Key result: [A singleton model is faithful and exactly compact-valued](#)
- Unital consequence: [Finite-dimensional algebra in the unital singleton case](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#) [9](#)

Level 0 (10 Cards)

Level 0

Representations without a unit-preservation requirement

Names $*$ -representations of a complex C^* -algebra on a Hilbert space when no unit equation is required.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Pure states as real extreme points

Defines purity inside the normalized positive state space.

`MathlibAnnex.Analysis.CStarAlgebra.IsPureState`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The GNS representation of a pure state is irreducible](#) · [A character extends to a pure state](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

An isolated point in a countable Baire space

Applies the Baire property to the closed cover by singletons.

`MathlibAnnex.Topology.exists_isOpen_singleton`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A unital singleton model acts in finite dimension](#) · [An isolated character away from the scalar character](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Maximal abelian *-subalgebras

Defines maximality by inclusion among commutative unital *-subalgebras.

`MathlibAnnex.Analysis.CStarAlgebra.IsMaximalAbelian`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A unital singleton model acts in finite dimension](#) · [A maximal abelian subalgebra containing a self-adjoint element](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A character separating a closed ideal

Finds one character which kills a closed ideal and detects a specified element outside it.

`MathlibAnnex.Analysis.CStarAlgebra.exists_character_annihilating_closedIdeal_of_not_mem`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Every operator in a singleton image is compact](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

The scalar character of the unitization

Names the character which reads the scalar coordinate and annihilates the embedded original algebra.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.infinityCharacter`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 0

The preimage ideal of the compact operators

Defines the two-sided ideal $J_\rho = \rho^{-1}(\mathcal{K}(H))$.

`MathlibAnnex.CStarAlgebra.compactPreimageIdeal`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Every operator in a singleton image is compact](#) · [A unital singleton model acts in finite dimension](#)

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Level 0

Full operator image in finite dimension

Constructs a preimage of an arbitrary operator by separately lifting its self-adjoint real and imaginary parts.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.surjective_of_irreducible_of_finiteDimensional`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Compact-operator conclusion for the unital singleton condition](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Singleton condition tested against all irreducible representations

Defines the unital singleton condition while allowing the quantified representations to be presented without a unit equation.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.IsSingletonIrreducibleModelAmongNonUnital`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Compact-operator conclusion for the unital singleton condition](#) · [Finite-dimensional algebra in the unital singleton case](#) · [Finite-dimensional representation space in the unital singleton case](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A compact-operator model for a representation

Packages faithfulness together with the equality $e(A) = \mathcal{K}(H)$.

`MathlibAnnex.Analysis.CStarAlgebra.IsCompactOperatorModel`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A singleton model is faithful and exactly compact-valued](#) · [Compact-operator conclusion for the unital singleton condition](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (6 Cards)

Level 1

Unitary equivalence of two representations

Defines equivalence by one surjective isometry intertwining all algebra actions.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Nonzero irreducibility without a unit assumption

Requires nonzero action as well as the absence of proper closed reducing subspaces.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The GNS representation of a pure state is irreducible

Applies the cyclic pure-state criterion to the space and vector constructed from the same state.

`MathlibAnnex.Analysis.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [A faithful singleton model forces simplicity](#) · [Faithfulness from a unique irreducible class](#) · [A character becomes a joint unit eigenvector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A character extends to a pure state

Chooses an extreme state in the compact face of all extensions of one prescribed character.

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_extension`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [A pure state detects a nonzero square](#) · [A character becomes a joint unit eigenvector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

An isolated character away from the scalar character

Finds an isolated point in the open complement of the distinguished scalar character.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_isolated_character_ne_infinity`

Immediate Card prerequisites: [An isolated point in a countable Baire space](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A maximal abelian subalgebra containing a self-adjoint element

Starts with the algebra generated by the given element and extends it by inclusion-maximality.

`MathlibAnnex.Analysis.CStarAlgebra.exists_maximalAbelian_containing_isSelfAdjoint`

Immediate Card prerequisites: [Maximal abelian *-subalgebras](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (3 Cards)

Level 2

A representative of the unique irreducible class

Defines when one nonzero irreducible $*$ -representation represents the sole unitary-equivalence class of such representations.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel`

Immediate Card prerequisites: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#) · [A character becomes a joint unit eigenvector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A pure state detects a nonzero square

Detects a nonzero element through its positive square without an algebra separability assumption.

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_nonzero_on_star_mul_self`

Immediate Card prerequisites: [A character extends to a pure state](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#) · [A unital singleton model acts in finite dimension](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A faithful singleton model forces simplicity

Uses irreducible representations annihilating proper ideals to rule out nonzero proper closed ideals.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.isSimpleCStarAlgebra_of_singleton_of_injective`

Immediate Card prerequisites: [The GNS representation of a pure state is irreducible](#)

Used by in this scope: [A unital singleton model acts in finite dimension](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (3 Cards)

Level 3

Faithfulness from a unique irreducible class

Uses a pure state on the unitization to detect any hypothetical nonzero element of the kernel.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.injective_of_singleton`

Immediate Card prerequisites: [A representative of the unique irreducible class](#) · [A pure state detects a nonzero square](#) · [The GNS representation of a pure state is irreducible](#)

Used by in this scope: [A projection represented by a rank-one operator](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

A character becomes a joint unit eigenvector

Transfers a normalized GNS eigenvector to the given representation through a unitary with a specified direction.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_unit_eigenvector_of_character_ne_infinity`

Immediate Card prerequisites: [A representative of the unique irreducible class](#) · [The GNS representation of a pure state is irreducible](#) · [A character extends to a pure state](#)

Used by in this scope: [The full character space is countable](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

A unital singleton model acts in finite dimension

Uses simplicity to put the identity operator in the compact image ideal.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton`

Immediate Card prerequisites: [Maximal abelian *-subalgebras](#) · [A faithful singleton model forces simplicity](#) · [A pure state detects a nonzero square](#) · [The preimage ideal of the compact operators](#) · [An isolated point in a countable Baire space](#)

Used by in this scope: [Compact-operator conclusion for the unital singleton condition](#) · [Finite-dimensional algebra in the unital singleton case](#) · [Finite-dimensional representation space in the unital singleton case](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (4 Cards)

Level 4

The full character space is countable

Uses separated unit eigenvectors to count the non-scalar characters, then restores the one scalar character.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.countable_characterSpace_of_nonUnital_singleton
```

Immediate Card prerequisites: [A character becomes a joint unit eigenvector](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4

Compact-operator conclusion for the unital singleton condition

Applies the nonunital compact-operator theorem to the same operator-valued map underlying a unital representation.

```
MathlibAnnex.Analysis.CStarAlgebra.Representation.faithful_and_compactOperatorModel_of_singleton_amongNonUnital
```

Immediate Card prerequisites: [A compact-operator model for a representation](#) · [Singleton condition tested against all irreducible representations](#) · [A unital singleton model acts in finite dimension](#) · [Full operator image in finite dimension](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 4

Finite-dimensional representation space in the unital singleton case

Applies the unital finite-dimensional-space theorem after checking the representation-interface conversion.

```
MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton_amongNonUnital
```

Immediate Card prerequisites: [Singleton condition tested against all irreducible representations](#) · [A unital singleton model acts in finite dimension](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 4

Finite-dimensional algebra in the unital singleton case

Distinguishes finite dimension of the representation space from finite dimension of the algebra embedded in its operators.

```
MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_algebra_of_singleton_amongNonUnital
```

Immediate Card prerequisites: [Singleton condition tested against all irreducible representations](#) · [A unital singleton model acts in finite dimension](#)

Used by in this scope: [An infinite-dimensional unital algebra has no separable singleton model](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5 (2 Cards)

Level 5

A nonzero projection with scalar corner

Constructs a projection in the original algebra from an isolated non-scalar character of a unitization subalgebra.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner`

Immediate Card prerequisites: [A maximal abelian subalgebra containing a self-adjoint element](#) · [An isolated character away from the scalar character](#) · [The full character space is countable](#)

Used by in this scope: [A projection represented by a rank-one operator](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5

An infinite-dimensional unital algebra has no separable singleton model

Uses the finite-dimensional algebra conclusion to contradict the proposed singleton condition.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.not_singleton_amongNonUnital_of_infiniteDimensional`

Immediate Card prerequisites: [Finite-dimensional algebra in the unital singleton case](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 6 (1 Card)

Level 6

A projection represented by a rank-one operator

Obtains all projection, normalization, image and compactness conditions on the same witnesses.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_rankOne_map`

Immediate Card prerequisites: [Faithfulness from a unique irreducible class](#) · [A nonzero projection with scalar corner](#)

Used by in this scope: [Every rank-one operator has an algebra preimage](#)

[Open Declaration Card](#) [Read exact source](#)

Level 7 (1 Card)

Level 7

Every rank-one operator has an algebra preimage

Turns one represented rank-one projection into every rank-one operator using dense orbits and a closed range.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_rankOne_of_singleton`

Immediate Card prerequisites: [A projection represented by a rank-one operator](#)

Used by in this scope: [Compact operators lie in a singleton representation range](#) · [Every operator in a singleton image is compact](#)

[Open Declaration Card](#) [Read exact source](#)

Level 8 (2 Cards)

Level 8

Compact operators lie in a singleton representation range

Obtains the inclusion of all compact operators in the image of the original algebra.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_of_compact_singleton`

Immediate Card prerequisites: [Every rank-one operator has an algebra preimage](#)

Used by in this scope: [A singleton model is faithful and exactly compact-valued](#)

[Open Declaration Card](#) [Read exact source](#)

Level 8

Every operator in a singleton image is compact

Excludes a noncompact image by making one separating character take both zero and one on the same projection.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_singleton`

Immediate Card prerequisites: [Every rank-one operator has an algebra preimage](#) · [A character separating a closed ideal](#) · [The preimage ideal of the compact operators](#)

Used by in this scope: [A singleton model is faithful and exactly compact-valued](#)

[Open Declaration Card](#) [Read exact source](#)

Level 9 (1 Card)

Level 9

A singleton model is faithful and exactly compact-valued

Combines faithfulness with both inclusions needed to identify the represented range.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.fairthful_and_compactOperatorModel_of_singleton`

Immediate Card prerequisites: [Compact operators lie in a singleton representation range](#) · [A compact-operator model for a representation](#) · [Every operator in a singleton image is compact](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Source exploration and boundary inputs

Full source declarations, helper paths and provider records are preserved separately from this selected Card reading scope.

[Explore 63 full source declarations](#) · [Earlier full-source Project PDF](#) · [Current selected reading PDF](#)

[Density below the continuum — source-only exploration](#). This preserved extension has 45 source declarations in its prerequisite closure and adds no Card to the selected 33.